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Critical Review of Processing and Classification Techniques for Images and Spectra in Microplastic Research

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Abstract

Microplastic research is a rapidly developing field, with urgent needs for high throughput and automated analysis techniques. We conducted a review covering image analysis from optical microscopy, scanning electron microscopy, fluorescence microscopy, and spectral analysis from Fourier transform infrared (FT-IR) spectroscopy, Raman spectroscopy, pyrolysis gas–chromatography mass–spectrometry, and energy dispersive X-ray spectroscopy. These techniques were commonly used to collect, process, and interpret data from microplastic samples. This review outlined and critiques current approaches for analysis steps in image processing (color, thresholding, particle quantification), spectral processing (background and baseline subtraction, smoothing and noise reduction, data transformation), image classification (reference libraries, morphology, color, and fluorescence intensity), and spectral classification (reference libraries, matching procedures, and best practices for developing in-house reference tools). We highlighted opportunities to advance microplastic data analysis and interpretation by (i) quantifying colors, shapes, sizes, and surface topologies with image analysis software, (ii) identifying threshold values of particle characteristics in images that distinguish plastic particles from other particles, (iii) advancing spectral processing and classification routines, (iv) creating and sharing robust spectral libraries, (v) conducting double blind and negative controls, (vi) sharing raw data and analysis code, and (vii) leveraging readily available data to develop machine learning classification models. We identified analytical needs that we could fill and developed supplementary information for a reference library of plastic images and spectra, a tutorial for basic image analysis, and a code to download images from peer reviewed literature. Our major findings were that research on microplastics was progressing toward the use of multiple analytical methods and increasingly incorporating chemical classification. We suggest that new and repurposed methods need to be developed for high throughput screening using a diversity of approaches and highlight machine learning as one potential avenue toward this capability.

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Introduction

Microplastics, plastic particles <5 mm in size,[1] have been increasingly recognized as a ubiquitous global pollutant.[2] Such findings have motivated rapid growth in microplastic pollution research over the past decade.[3] This has led to the development and diversification of methods to quantify and characterize microplastics found in environmental matrices, including air,[4] water,[5] biota,[6] and sediments.[7] The result is a wide range of sampling, sample processing, sample analysis, data analysis, and reporting methodologies, which complicate cross-study comparability and larger scale synthesis, and confuse an expanding base of new practitioners.[8] Sample and data analysis have become critical bottlenecks in microplastic research schemes because individual samples of environmental media can contain hundreds to thousands of microplastic particles,[9] and analysis procedures can be incredibly time-consuming and laborious.[10] The techniques employed in microplastic research have a wide range of costs (time and monetary) and benefits (information quality).[11] Scientists responding to the questions of managers and policy makers need the highest quality information with the least cost. As the field expands and we learn more about this complex contaminant suite, experienced practitioners must address these growing pains by evaluating best analytical practices with a focus on high throughput and automated techniques.

Microplastic analysis often includes images and spectra (e.g., spectroscopy, spectrometry) to quantify and classify plastics. Images are pixel maps with one to three information bands (e.g., monochrome image = one band, RBG image = three bands). Images are not generally able to discriminate chemical properties without many additional steps to prepare the sample for imaging. Spectral analysis strategies can produce greater than 100 bands of information (e.g., Raman spectroscopy, infrared (IR) spectroscopy). Spectral images can be created by collecting spectra along a surface at multiple locations. The increased information of spectral techniques allows polymers and sometimes additives to be clearly identified, which can be utilized for microplastic characterization and chemical identification (referred to here as chemical or spectral techniques).[9,12] Similarly, multispectral techniques incorporate some number of signature bands between 3 and 100. In many fields, multispectral imaging provides a balance between data quality and speed of acquisition for classification strategies.[13] However, the use of multispectral strategies in microplastic research is yet to be realized. After an image or spectral data is collected, it needs to be processed and classified. Data processing includes everything that happens to the data to prepare it to be classified. Data classification happens after data processing and is the action of discriminating whether a particle is a plastic or not and what type of plastic it is. A critical analysis of microplastic analytical techniques involved in image and spectral processing and classification would aid the development of future high throughput techniques.

Here, we review 127 peer-reviewed publications to critically assess the state of data analysis in microplastic research (Methods, Supplemental Material). First, we tabulate and discuss the reviewed study locations, foci, and methods. Closely connected to this review is that of Primpke et al.,[10] which focuses on the hardware and methods involved in collecting quality data on plastics. This review outlines the current state of image processing, spectral processing, image classification, and spectral classification. We also discuss data analysis quality assurance and quality control (QA/QC) procedures, and data storage and sharing practices. In each section, we describe opportunities for the field to advance and describe critical methods that should be reported for reproducing and comparing the study with other studies.[8] In the cases of reproducible methods reporting, image and spectral reference libraries, and image analysis tutorials, we also created tools to fill critical data processing and classification needs and provide them in the supplementary information. We end by discussing the paths toward future developments that address the sample and data analysis bottleneck through automation and high throughput techniques.

General Diagnostics of Reviewed Papers

The 127 journal articles assessed in this review were published between 1995 and 2019 and were conducted in every major region of the globe except Antarctica and many of the Pacific Islands. These included field-based and experimental studies focused on lakes, streams, marine, and terrestrial environments in surface water, water column, sediment, biota, wastewater, and table salt. We reviewed research articles, methods papers, opinions, and other review articles to get a wide perspective of the field. The most common chemical (spectral) technique was Fourier transform infrared spectroscopy (FT-IR, a common type of IR spectroscopy), and optical microscopy was the most common visual technique (compatible with images) (Fig. 1).

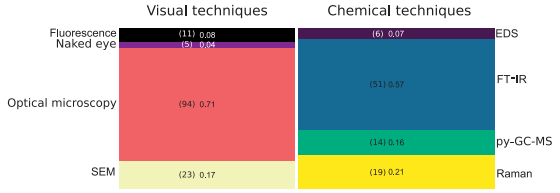


Figure 1. The number of times each method was used in the reviewed publications in parentheses followed by the proportion of the total methods used within the visual and chemical technique groups. FT-IR and optical microscopy were the most common techniques employed for microplastic classification.

Data Processing

Image Processing

Once an image is taken, there are many software choices available (e.g., Image J, Simple ITK) to correct the color, threshold (set the limits that differentiate background from particles), and measure the size and shape of the plastics to prepare it for classification. Optical microscopy, scanning electron microscopy (SEM), and fluorescence microscopy each have image processing steps. When spectral maps are fully classified (e.g., Raman and IR spectral maps), their data are organized as a discrete labeled image and can be analyzed similarly to other images,[12] which is a post-processing step of spectroscopy. We also discuss some manual characterization strategies currently used to identify opportunities for image automation. Each processing step affects the quality of the information used for classification of the suspected plastic particles, which we discuss in the Image Classification section.

Color. Color of microplastics can be determined via photography in the visible spectrum. Microplastics are found in a wide variety of colors and patterns.[14] Color is usually described qualitatively. Most studies create a small subset of common color categories in which to bin their suspected microplastics (e.g., black, red, blue).[14,15] However, Robards et al.[16] employed an elaborate color characterization scheme using a Munsell Notation color scheme of hue, value, and chroma. The previous examples were that of manual color characterization, which is the most common mode of characterizing color. To automate image processing, more quantitative colors can be characterized. For example, Maes et al.[17] used RGB values to create a quantitative index for fluorescence values $(R + G)/R$.

Thresholding. Thresholding is an image processing approach for extracting objects of interest from the image background using differences between the background and object characteristics, which has been applied to extract microplastics using color (Fig. 2), shape,[12] and fluorescence intensity.[18,19] Thresholding is a mandatory step in the automation of particle identification and characterization. A distinct advantage of any thresholding routine is that the plastic particles do not need to be completely extracted physically from the sample for classification. For example, the practitioner can automatically locate microplastics in an image using the Threshold by Color tool in Image J (Fig. 2) if a standardized background with different color characteristics than the study microplastics is used. Primpke et al.[12] thresholded images of FT-IR focal plane array outputs by shape to quantify fibers. Plastic fibers were discriminated from other shaped particles by quantitative shape factors (elongation factor and roundness). In fluorescence microscopy, fluorescence intensity is a function of the instrument parameters (light source intensity, light source angle, camera acquisition capabilities), ambient light, and background fluorescence (described thoroughly in Primpke et al.[10]). Once the optimal fluorescence intensity, represented in grayscale,[18] or color index,[17] is identified for a standardized setting, it can be repeated to automate particle thresholding. After thresholding, there are typically artifacts (e.g., background interference, noise, and non-uniform illumination) that must be manually inspected and corrected to refine thresholding routines.

Particle Size, Shape, and Quantity. Image analysis is used in microplastic research to rapidly measure the size,[9,20–22] shape,[12,23–25] and quantity,[9,26,27] of plastic particles. First, image analysis software gives the dimensions of the image in physical units. Some imaging systems can interpret such metadata to produce a scale within the image, while others need a size calibration standard such as a graded ruler placed within the image window. The scale bar or calibration standard can then be used to scale measurements of the image by determining a conversion rate between the pixel length and the scale units (e.g., mm) (Fig. 2). If an image is thresholded, these measurements can be automated (Automated Particle Analysis, Supplemental Material),[28] otherwise manual measuring procedures are also possible.

Particle size was most typically measured as the Feret maximum,[25,29] average of the Feret minimum and maximum,[30] or projected surface area.[31] Feret dimensions are defined by the smallest hole an object can fit through and the smallest hole an object could fit through regardless of orientation. Exact size dimensions of particles can aid in fate and transport analysis and modeling as well as risk assessment.[32,33] For fibers, the total length (long axis) of the fiber can be traced using straight or segmented lines, which may be useful for calculating volume since the width of fibers is relatively symmetric.[23] Otherwise, a thresholded image can be processed with a skeletonize function to reduce the fiber to the centerline with a width of one pixel for length calculation.[12]

Particle shape can be explicitly described using measured values (e.g., aspect ratio, form factor, roundness, convexity)[24,34] or qualitatively described (e.g., linear,

(a)



(b)



(c)

ParticleID	Area	Perim.	Circ.	Feret	FeretX	FeretY	FeretAngle	MinFeret	AR	Round	Mass	Material
1	1269.79	165.50	0.58	58.33	95.14	50.14	39.86	34.67	1.77	0.56	0.16	Foam
2	1409.59	166.25	0.64	52.45	138.19	23.58	8.59	41.40	1.18	0.85	0.19	Foam
3	561.50	141.50	0.35	40.93	9.63	37.83	42.56	26.75	1.54	0.65	0.10	Foam
4	940.69	130.54	0.69	42.41	52.98	16.34	141.86	31.86	1.23	0.81	0.26	Foam
5	394.50	96.01	0.54	33.86	6.42	78.05	3.28	18.45	1.78	0.56	0.05	Foam
6	284.47	85.00	0.49	24.04	88.42	80.66	20.91	17.72	1.16	0.86	0.39	Foam
7	256.95	82.03	0.48	30.10	47.23	75.14	170.58	10.03	3.42	0.29	0.20	Foam
8	114.04	46.20	0.67	15.68	126.03	85.36	80.13	10.60	1.56	0.64	0.01	Foam
9	137.09	54.95	0.57	20.85	148.86	78.57	3.28	9.40	2.27	0.44	0.02	Foam
10	6.18	10.04	0.77	3.09	127.37	102.75	142.85	2.73	1.07	0.93	0.00	Foam

Figure 2. Thresholding and setting the scale of the image (a) to extract the plastics from the background using Threshold by Color in Image J (b) to semi-automate measurement of particle count, shapes, lengths (c). Photo credit Hannah Hapich and Shannon Tarby (Gray Lab, University of California, Riverside).

spherical).[14,35] Most microplastic studies use qualitative shape classifications. Current microplastic shapes common in most studies include fibers, fiber bundles, fragments, foams, films, spheres, and pellets.[14] These classifications have quantifiable shape factors that are inherent in their interpretation but also rely heavily on surface topology, size, thickness, color, and compressibility.[14] For example, a foam and a pellet can both be spherical and similar in size, but foams are typically white and pliable, and pellets are hard and do not have a standard color.[14] Within our review, Cole[24] included a quantitative threshold for morphological classification, stating fibers must have a minimum aspect ratio of 3:1 (length:diameter) and Primpke et al.[12] defined an elongation factor (>3) and roundness limits to fiber shapes.

Particle count by size class (e.g., 1–5 mm for mesoplastics) is the most commonly used technique to quantify microplastic.[36] Particle count alone is not fitting for all purposes in microplastic research and should be reported together with the particle mass and multiple size and shape descriptors per particle if possible. Mass is especially important for mass balance calculations and modeling.[37] In a study by Hermabessiere et al.[23] particle volume was used as a surrogate for estimating mass from particle dimensions. Additionally, Simon et al.[38] provided an approach to estimate plastic mass based on particle counts using FT-IR imaging of particle volume and polymer density.

Opportunities for Image Processing Development. Image processing software was underutilized in the reviewed studies, which relied heavily on manual techniques. However, most studies shared images that could be processed with particle analysis software (Review Metadata, Supplemental Material), highlighting the applicability of the technology to this field. Since many of the reviewed publications used manual approaches without thresholding routines, we provide a tutorial on the application of color thresholding for measuring the color, size, and shape of all plastic particles in the Supplementary Information.[28] Automatically quantifying all particles within images by color, shape, surface topology,

and size using image processing software reduces ambiguity while increasing reproducibility and applicability of microplastic research. It is important to note that filters on the camera and image alterations will change the color of the microplastic and proper setup (white light illumination, white correction, and controls on color) should be considered whenever highly accurate color characterization is desired. Measuring particle by particle projected surface area or long and short axis could narrow the range of error in mass balance assessments of microplastic abundance[37] and contribute to the quantification of risk (which is influenced by microplastic particle size and count).[39] Quantitatively characterizing shape of all particles could be used to define limits for spherical objects based on circularity or roundness factors, similar to how fibers have a distinction based on aspect ratio. No standard scheme for the description of microplastic surface topology (described in depth in the Image Classification section) currently exists, but the density or size distribution of surface features such as pits or cracks, or indices of smoothness/roughness may provide useful information on plastic supply, transport, and/or weathering.[40–44] Lastly, when reporting image processing steps there should be specific discussion of any color processing, thresholding, and size shape and quantification parameters.[8] These factors were rarely mentioned in the reviewed literature.

Spectra Processing

Before any spectral data from IR spectroscopy, Raman, pyrolysis–gas chromatography–mass spectrometry (Py-GC–MS), and energy-dispersive X-ray spectroscopy (EDS) can be used to identify plastic types and sort plastic from nonplastic materials, the data must go through processing. The aim of the data processing step is to eliminate or reduce the contributions of undesirable signals that may occur while obtaining spectra. The contributions of undesirable signals may reduce the quality or reliability of analysis results, and thus must be removed or minimized where possible. Many spectroscopy instruments will automate this step, but it is important to understand, interpret, and report the techniques used. For example, Fig. 3 shows three Raman spectra of polypropylene that were acquired from different samples using different systems in different labs and underwent different data processing steps, the spectra are very different. The most commonly used forms of spectral processing methods include baseline and background subtraction, smoothing, and data transformation. Py-GC–MS and EDS do not commonly have intensive data processing steps besides removing background signals. To the best of our knowledge, there is no perfect data processing strategy that works for every spectrum. All have their own benefits and drawbacks, and preferred methods are highly dependent on the application and the sample.[45]

Background and Baseline Subtraction

Background and baseline contributions in spectral signatures result from instrumental setups, measurement modes, ambient conditions, and the materials in the sample matrix and mount. Baseline and background correction are commonly applied via instrumental, experimental, and computational methods.[47–49] We define background as signal interference occurring due to anything external to the sample and baseline due to the sample itself.

Baseline. In Raman spectroscopy, the main sources of baselines (Figs. 3 and 4) are fluorescence, while the source of baseline in IR can be absorption from carbon black and scattering effects. Sources of fluorescence include additives such as pigments and plasticizers, and impurities such as biological material, biofilms, and degradation products.[50] Fluorescence signals, when strong, can overwhelm Raman bands.[51–53] Depending on the sample, it is possible to avoid fluorescence signals by selecting a laser that does not cause them—very often this is at an near infrared (NIR) wavelength, such as 785 nm. Confocal Raman microscopes can suppress fluorescence signals by employing high numerical aperture objective lens and a confocal hole. In IR spectroscopy, strong light absorbing compounds such as carbon black may drastically distort the baseline which has been observed in tire wear particles.[54] Additionally, IR baseline can be due to scattering of IR light which is due to the surface roughness of the sample.

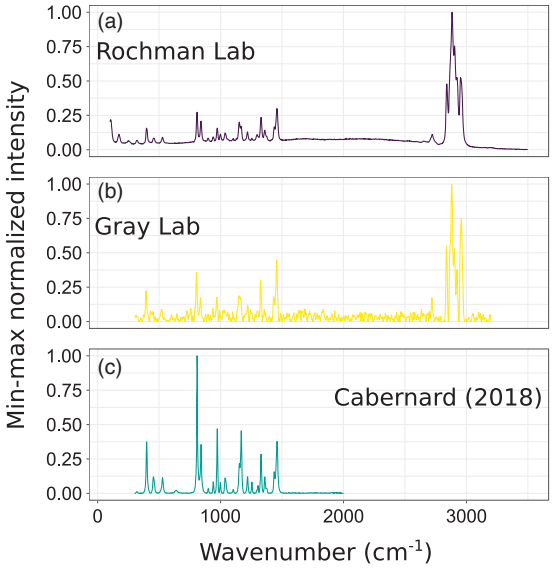


Figure 3. Three Raman spectra of polypropylene at different stages of processing. They were acquired using different systems in different labs. (a) Unprocessed polypropylene spectrum from the Rochman Lab, (b) baseline corrected, not smoothed polypropylene spectrum from the Gray Lab, (c) baseline corrected and smoothed spectrum from Cabernard et al.[46]

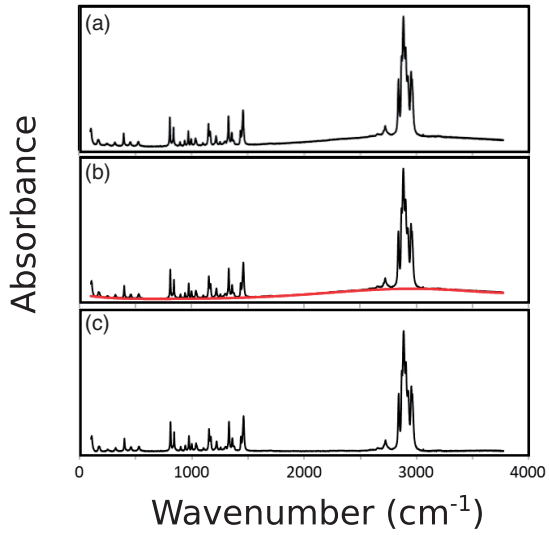


Figure 4. (a) A Raman spectrum of polypropylene. Note that baseline increases gradually toward the high Raman shift. (b) The baseline is fitted with a sixth-order polynomial curve. (c) The baseline corrected spectrum of polypropylene.

Common methods to remove baseline include offset correction (subtracting a constant), fitting, and subtracting one of more linear or polynomial function(s), or more complex tools such as rubber band polynomial baseline fitting techniques.[55] The process of baseline correction can be done manually[51] or automated with routines that adaptively fit to the baseline.[56–59] In general, these correction techniques are applicable to Raman and IR spectroscopy.[60] Figure 4 shows an example of baseline correction of a Raman spectrum using polynomial curve fitting.

In Py-GC–MS and EDS, baselines from other elements and chemicals within samples can result in the inability to identify target chemicals and elements below specific mass percent thresholds. Higher sensitivity instruments and alternative techniques are needed to detect compounds of interest when the mass is low.

Background. A background can occur in attenuated total reflection (ATR) FT-IR spectroscopy due to difference in the applied pressure on the crystal or caused by materials and atmospheres present in the instrument compartments, e.g., water vapor and carbon dioxide. For background correction in IR spectroscopy, the contributions of the sample to a spectrum must be separated from the contributions of other sources. A background spectrum is typically measured in absence of the sample and subsequently subtracted from the measured sample spectrum.[61] This step is often directly performed within the measurement software as part of the data treatment prior to output of the resulting spectrum.[62] However, if this process is not 100% successful, further techniques exist. For example, Primpke et al.[9] removed the remaining atmospheric CO_2 absorbance by replacing this range within the spectrum with a straight line. However, contributions from moisture and carbon dioxide are minimal in Raman spectroscopy.

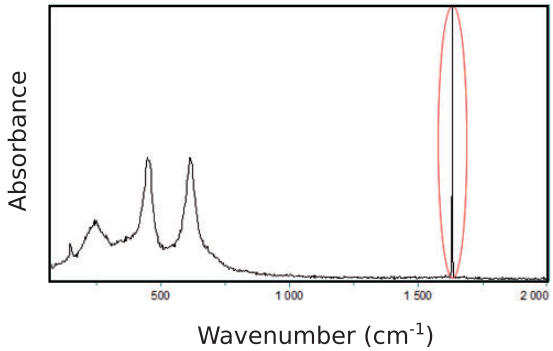


Figure 5. A Raman spectrum of titanium dioxide (TiO_2, rutile) with a cosmic ray (also known as a spike) marked in red.

Background signal in Raman spectroscopy is usually related to the wavelength of the laser and detector materials used in the instrument or cosmic rays. As the spectral range of lasers encompasses UV (e.g., 244 nm) to near-IR (e.g., 785 nm), instruments may use optics and detectors of different materials (e.g., silver mirrors for visible range versus aluminum (Al) mirrors for UV or near-IR range; silicon (Si) detectors up to 1000 nm wavelength versus indium–gallium–arsenide detectors for 800 nm and up). The instrumental background is corrected for the variance in reflectance or transmittance of optics and quantum efficiency of the detector across the spectral range. Correction for the instrument background uses standard light sources or standard reference materials whose wavelength dependences are carefully characterized and recorded as a reference to calculate a correction factor.[63] Spectra acquired from the instrument are then corrected using the correction factor. In Raman spectroscopy, there is one more source of background signal, cosmic rays. Cosmic rays are very sharp, with strong peaks that appear at random when these high energy particles interact with the detector. It is relatively easy to recognize a cosmic ray signal because it is very narrow and high intensity (see Fig. 5). The best way to correct for cosmic rays is to acquire and compare multiple spectral measurements of the same material. Raman bands are consistent, while cosmic rays appear at random. The correction process removes spikes that appear in one measurement but not in others.

We are not aware of background interference in Py-GC–MS but in EDS measurements, background from adjacent surfaces to the sample location can contaminate the sample spectrum. In these cases, background correction can be achieved by measuring the adjacent signal and subtracting it from the sample spectra.

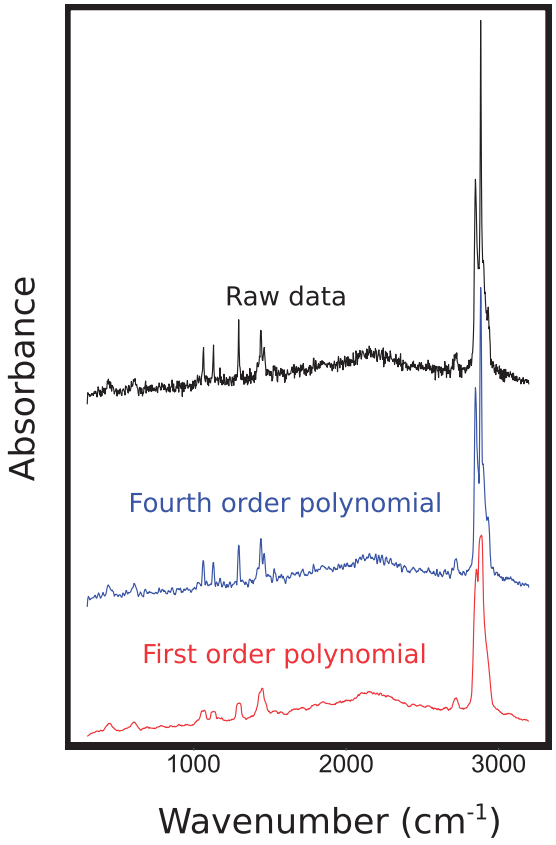


Figure 6. Example of a high-density polyethylene (HDPE) Raman spectrum unprocessed (black) and after smoothing using a fourth (blue) and first (red) order Savitzky–Golay filter of 12 points. The fourth-order polynomial decreases noise in the 2000 wavelength range without changing the signal shape or intensity, while the first-order polynomial drastically changes the shape and intensity of the signal peaks at wavenumbers ~2800 and ~1500.

Smoothing and Noise Reduction

For the analysis of a spectrum, the signal-to-noise ratio (S/N) is an important indicator for spectral quality. S/N is a function of many factors including the sampling volume, the concentration of target materials in the sample, the instrument setup, power of the light source, signal acquisition time, etc. (covered in depth in Primpke et al.[10] and Löder and Gerdts[64]). It is ideal to achieve a high S/N by acquiring a strong signal. When it is not possible or practical, smoothing or extracting regions of high S/N can improve S/N by reducing noise.

Smoothing is used to decrease the noise by reducing small variations at high frequency in the spectrum while maintaining large variations at low frequency. Spectral smoothing is performed in Raman (Figs. 3 and 6) and IR spectroscopy. The most common smoothing technique used in the reviewed studies was the Savitzky–Golay filtering method.[51,65] Here, the data are replaced with a running bin polynomial function. Other smoothing methods commonly employed in spectroscopy are Kernel smoothing, LOWESS, and Gaussian Blurring.[66] Smoothing should be used with caution, as it has some disadvantages for the spectral information (Fig. 6). If a spectrum is over smoothed, it can degrade the resolution and artificially broaden the peak width. Also, peak positions, and shapes can be distorted, and subtle spectral features may be lost. In Fig. 6, the shoulder of the peak around 1500 cm^{-1} in the unprocessed spectrum is lost in the first-order smoothed spectrum.

In Raman systems, detector response limitation can also introduce noise. A charge-coupled device (CCD) is a Si-based detector, and its response starts to decline ~900 nm. For over ~950 nm, the decline in the response is significant. This results in decreased S/N over ~2000 cm^{-1} Raman shift region when recording a Raman spectrum using a 785 nm laser. CCD responses also vary as a function of wavelength (CCD Figure, Supplemental Material). For this reason, it is not uncommon to exclude spectral range 2000 cm^{-1} and higher Raman shift from analysis (Fig. 3).[46,67]

Data Transformation

The spectral signal intensity is dependent on many parameters including: instrumental, experiment conditions, system components such as detectors (Primpke et al.[10]) and environmental conditions. Data transformation is common in FT-IR and Raman spectroscopy. To perform library searches or use multivariate tools, it may be necessary to perform data transformation to normalize the intensity scales and accentuate signal peaks between reference spectra and the sample spectrum, or between sample spectra. In the reviewed studies, max–min normalization,[9] vector normalization,[68] and derivatives were used.[68,69] Max–min normalization scales intensities to unit height (0,1) by subtracting each intensity by the minimum intensity, and then divides by the range (max–min) of intensities. Vector normalization treats intensities as a vector and scales them to the unit norm (=1) by calculating the square root of the sum of intensity squares. The first and the second derivatives remove elements of baseline, accentuate shoulders, and differentiate composite bands. Because derivatives can be calculated using the Savitzky–Golay filtering method also used for smoothing, the first and the second derivative calculation can smooth the spectrum too. Figure 7 shows an extraction (700 cm^{-1}–1600 cm^{-1} region) of a Raman spectrum of polypropylene and its derivatives. The peak at ~1035 cm^{-1} has a shoulder ~1045 cm^{-1}. The shoulder is more separated and easier to characterize than the original spectrum when using the first derivative (Fig. 7b) and best separated in the second derivative (Fig. 7c).

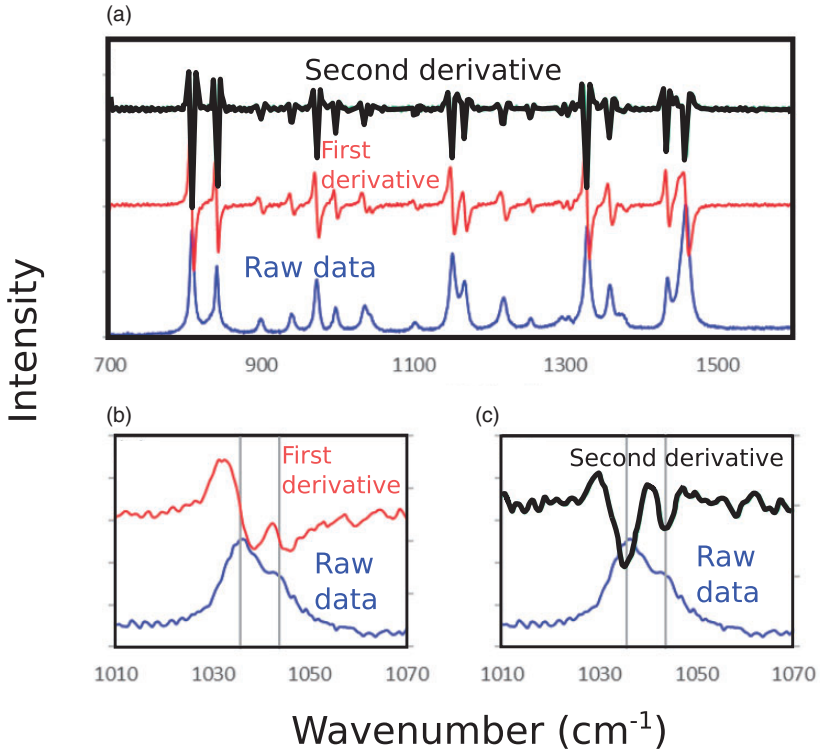


Figure 7. (a) An extraction (700 cm^{-1} to 1600 cm^{-1} region) of a Raman spectrum (blue) of polypropylene and its first (red) and second (black) derivative. (b) and (c) Zoomed to 1010 cm^{-1} to 1070 cm^{-1} region where the first and the second derivative separate the shoulder at ~1045 cm^{-1} from the main peak at ~1035 cm^{-1}.

Opportunities for Spectra Data Processing Development

Advanced intensity correction algorithms and new plastic specific software development should be considered, and all processing steps should be adequately described in research reports. To correct the instrumental spectral sensitivity in Raman spectroscopy, the intensity correction system (ICS) can be used. An ICS should be used for two reasons: to correct waves and background noise received in a spectrum with high background when edge filter or notch filters are used, and to obtain an identical relative peak intensity and spectrum shape when different lasers and gratings are used. IR spectroscopy has similar procedures for standardizing between transmission and reflectance spectra that has yet to be realized in the reviewed literature.

We suggest developing advanced processing tools specific to plastic, as has been done in other fields.[50] Using multivariate analytical tools, patterns in the data can be modeled and used to predict the newly acquired data of a similar type.[55,66,70] Some of the advanced methods for data processing used in Raman and IR spectroscopy include principal component analysis, linear discriminant analysis, multiple linear regression cluster analysis, and partial least squares regression. A few open access tools developed specifically for plastic that are available now are Open Specy[71] and siMPle.[72]

In publications using spectral data analyses, smoothing, baseline correction, data transformation, and background subtraction processing steps should be discussed. In this review, very few of the papers that used spectroscopy mentioned any of these topics, much less discussed them in the level of detail required to support reproducibility.[8] Some studies reported that they used a software to analyze the spectral data, but did not report whether their analysis incorporated processing or not.[73–75] Other studies stated which software they used to remove background signals,[29] but not the specific algorithm or parameters used. These facts make developing best practices for processing plastic spectra challenging. Moving forward, authors should be encouraged to share details and access to algorithms used for this purpose as provided by Ghosal et al.[59] There should also be a thorough discussion of which processing parameters work best in which situations when parameters are tuned to the most optimum S/N ratio for each particle.

Data Classification

Image Classification

Visual classification is typically the initial step in distinguishing microplastic particles from non-synthetic microparticles.[76] Physical manipulation of the sample with tweezers and an optical microscope is one of the most common

plastic classification steps, but this technique has produced considerable barriers to sample throughput that automated image classification may be able to solve. Visual classification cannot describe polymer types but gives an indication as to whether a particle is a plastic or not. Visual inspection is subject to a high probability of false positive results due to the bright color and plastic-like appearance of some natural materials (e.g., minerals and microorganisms), and false negative results due to biofilm formation concealing the view of the plastic surface. Currently, visual inspection alone is rejected as a microplastic classification technique by most reviewers and spectral classification (discussed in the next section) of at least a subset of the suspected plastic is required.[77] Standardization and inter-laboratory comparisons for microplastic visual classification techniques are ongoing.[78] While these methods are currently developing, we describe the reference libraries, morphology, color, and fluorescence intensity considerations that researchers use to identify microplastics from environmental samples. Using multiple techniques and settings (e.g., lighting and background magnification) and incorporating several lines of evidence will aid in the accuracy of visual microplastic classification.[14,76,79]

Reference Libraries. Reference libraries allow quick retrieval of microplastic shapes, surface topologies, colors, and fluorescence to validate a visual classification of microplastics. Right now, there exists no standardized reference library for identifying microplastics visually. Most labs using visual techniques are self-trained, which may cause variance in results, as evidenced by investigations finding only 39% to 68%[51,80] of suspected microplastics correctly identified as plastic. Taxonomic classifications have been proposed but many researchers are still choosing to use their own classifications and not reporting the definitions of their taxonomic groups, which limits comparability between studies.[14,35,81] Keys exist for distinguishing microplastics from other non-plastic microparticles, and subsequently categorizing microplastics into morphological groups.[35,14] Many papers share images of their plastic particles which should serve as a useful reference for other researchers[82–90]; however, these reference images are typically limited in number and diversity and are difficult to compile from the literature. Recognizing a need, we created an R code to download images from the Elsevier journal database API (Elsevier API R, Supplemental Material),[91] share images from UC Riverside's database (Images, Supplemental Material),[92] and outline which journal articles from our review share imagery of microplastics (Review Metadata, Supplemental Material).

Morphology: Shape and Surface Topology

Microplastics exist in a wide array of unique shapes and surface topologies different from those commonly found in the natural environment, which can aid in their classification from environmental samples. Shape is a high order of classification which considers the object's dimensions from a scale where the entire object is in view. Surface topology is a finer scale description of microplastic where the surface is inspected at a scale zoomed onto the surface of the object (typically $> 10\times$). There is not a common nomenclature for surface topologies nor has there been an in-depth discussion of their use. As a reminder, microplastics are commonly described by morphological taxa that include fibers, fiber bundles, fragments, foams, films, spheres, and pellets.[14] These taxa are composed of combinations of shapes, surface topologies, sizes, and colors. Some of these shapes and surface topologies are easy to distinguish from natural particles, while others are not.

Unique shapes such as microbeads, pellets, and fibers are less common in nature than fragments which describe any irregular shape. Microbeads, pellets, and foams are often quite distinct from natural particles due to their symmetric spherical shape which is uncommon in nature. However, coal fly ash can be symmetrically spherical and look similar to microbeads, and further chemical analysis, such as EDS, is needed to distinguish coal fly ash from microplastics (Fig. 8).[93] Fibers are elongated linear objects (often derived from clothing) and their shape stands out in environmental samples, but there are also many non-anthropogenic elongated fiber shaped objects such as plant matter (Fig. 8). Anthropogenic fibers can be distinguished from non-anthropogenic fibers based on anthropogenic fiber's consistent thickness and color throughout (Fig. 9c2).[94]

Surface topology is often assessed with SEM or microscopy.[29,84,97–100] Surface topology of plastic-like objects at high-magnification in the micro- to nanometer range is often assessed with scanning/backscattered electron microscopy (SEM-BSEM), following visual prescreening by optical microscopy.[74,82,101,102] Unique surface features that have been associated with plastic include cracks, pits, fractures, flakes, and granular oxidation, mainly produced by mechanical and/or oxidative weathering in the environment.[62,74,103,104] Most of these morphological modifications increase particle surface area, promoting the adsorption of chemicals such as persistent organic pollutants and/or pigments, as well as the adhesion of debris, biofilms, and microorganisms on the surface of plastics.[101,105,106] Although the adhering objects are usually distinguishable based on their brighter SEM-BSEM contrasts caused by charging and/or inorganic elements, their non-plastic nature can further be investigated in an SEM by EDS, as explained later in the Spectral Classification section.[101]

Surface enhanced microscopy and optical microscopy are used to describe the surface topology of the different taxa of microplastics including fibers, films, beads, foams, and fragments (Figs. 8 and 9).[61,67,74,95,106] Freshly

manufactured plastics typically have a smooth surface, while weathering roughens the surface (Figs. 9a2 and 9c2).[24,84,93] Fibers and films have many non-anthropogenic shape and color analogues such as hair, roots, or insect body structures. These types of particles may be examined at higher magnification, which aids in the elimination of some natural materials by visualizing cellular structures.[107] Hair can be eliminated on the basis of scaly surface topology. However, insect body structures can also have smooth surfaces and look similar to films which may require further classification steps using shape, surface topology, and color.[29] In optical microscopy, expanded polystyrene foam plastic can have unique surface bubbles with glossy rainbow streaks (Fig. 9d2). In SEM images at higher magnification, other foam plastic surface topology shows unique micro-holes that can become larger under environmental exposure, resulting in a rougher surface.[95] Hard plastic fragments, mostly derived from larger plastics, can display surface cracks, pitting, and streaking (Fig. 9b2) with morphologies that are potentially diagnostic,[102] although surface topology has yet to be quantified and critically described.[103,106]

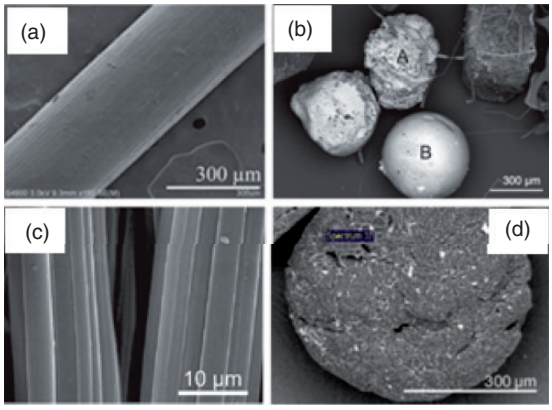


Figure 8. Examples of possible false positive plastic classification in environmental samples. (a,b) SEM images showing fibers with similar surface topology, but of different type (a) PET identified by IR and (b) viscose. (c,d) SEM images of optically resembling plastic-like objects, confirmed as (c) coal ash and (d) polymer fragment by EDS based on their inorganic and polymeric composition, respectively. Adapted from Eriksen et al.,[93] Ding et al.,[95] and Kramar et al.[96] with permission from The Royal Society of Chemistry, Springer Nature, and Elsevier, respectively.

Color and Appearance

Color and appearance have been used to identify sources of contamination in samples,[108] and distinguish plastic from non-plastic particles.[98,109,110] Conflicting views exist regarding whether microplastics are homogeneously colored and matte,[21,111] or patterned and shiny.[14,89] Because shininess is determined by the light source, the intrinsic reflectance of the material and any liquids or other materials sorbed to it, we are hesitant to include suspected microplastics based solely on shininess of the microparticles without more information. Another common color consideration is the idea that plastics are brightly colored,[89] as shown in Fig. 9, many plastics are muted or dull. Additionally, many naturally occurring microparticles exist in bright colors. For example, minerals, algae, and some plant material can be brightly colored. Dris et al.[112] recommends considering factors beyond color for green or transparent particles due to the prevalence of these colors in nature. Additionally, there is likely a bias against clear microplastic particles because they are difficult to see with light microscopy.[113] Some studies proclaimed plastic particles as having an unnatural color,[114] but this definition is not descriptive

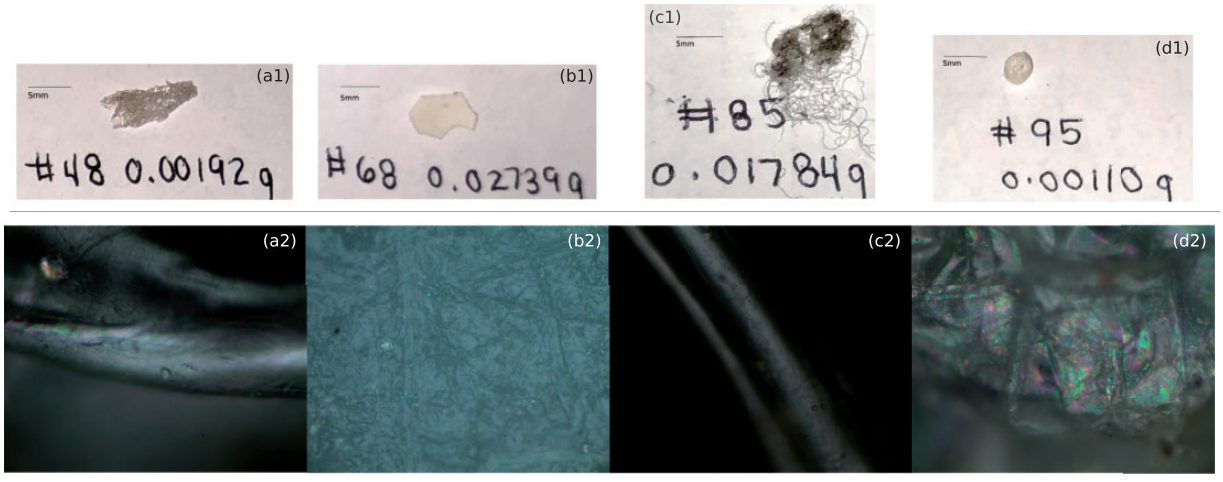


Figure 9. (a1) Film, (b1) hard fragment, (c1) fiber, and (d1) foam. Images (a1) to (d1) were taken with a camera phone from 60.96 cm (24 in.) away. Images (a2) to (d2) are corresponding surface textures taken with a 50× objective microscopy camera.

and does not incorporate the potential for plastics that are dyed to mimic nature. Also, natural fibers may be dyed and appear synthetic.[14] For example, cotton or wool fibers can appear brightly colored and may not be easily distinguished from plastics based on color. For the numerous reasons listed above, we declare categorical color descriptions as used today, a relatively ineffective technique for distinguishing microplastic particles from natural particles.

Fluorescence Intensity

In microplastic classification, fluorescent dyes provide another approach for the identification of potential plastic particles. Samples are treated with hydrophobic dye (e.g., Nile red) that clings to plastic and then imaged under magnification with illumination with specific wavelengths of light that cause the dye to fluoresce. Typically, a minimum (threshold) fluorescence intensity value is set to distinguish plastic from non-plastic. Maes et al.[17] developed a fluorescence index based on the RGB scale of their Nile red images which they used to set a cutoff intensity for positive plastic classification and were able to distinguish polymer types based on the intensity. Erni-Cassola et al.[18] used monochrome fluorescence to set a cutoff intensity. It is important to remember that these intensity values are relative and will need to be calibrated to any device's settings and exterior lighting conditions.

Opportunities for Visual Classification

We have only scratched the surface of what image analysis and classification can accomplish.[19] For example, an accurate quantification of colors could lead to quantification of the unique colors that researchers use to identify plastics and may allow color thresholding of visible images of samples for rapid analysis and quantification.[29,114] We recommend identifying multiple unique quantifiable descriptors for shape, color, size, surface topology, and fluorescence properties that can be used to classify plastic particles and develop multispectral classification techniques. For example, by using information obtained from fluorescence, visual, reflectance and other microscopy techniques in combination, researchers may be able to find unique combinations of information across the bands and advance the ability to identify plastics in heterogeneous mixtures. Using particle analysis to automate the quantification of these factors as described in the Image Processing section could greatly speed up the process. Many of the studies we reviewed shared images to illustrate morphological distinctions (Review Metadata, Supplemental Material). This wealth of microplastic imagery could be used today to identify quantitative thresholds unique to plastics by extracting particle color, morphology, and fluorescence information and applying machine learning tools. Machine learning procedures currently being developed are described later in the Machine Learning Applications section. These automated processes could be used to prescreen samples for plastics and could be cheaper, less time intensive, and easier to deploy. To aid these developments, future publications should thoroughly describe the morphological, color, or fluorescence features that are applied to distinguish plastics from non-plastics using images when possible. For example, one can use an image to quickly convey what they mean by shiny or pitted.[21,102,111]

Spectra Classification

Following or in lieu of visual classification, samples can be analyzed for spectral identity. If visual classification has been conducted, it is required that at least a subsample of the suspected particles are analyzed for spectral identity.[110,115] Spectral analysis of microplastic particles is currently a time consuming and costly endeavor.[46,116] Using spectral analysis methods, one can obtain many of the chemical components that make up an unknown sample (polymer, pigments, additives, etc.). Reference library searching software can provide an array of reference spectra and matching procedures with which the sample spectrum can be classified. However, many studies still choose to develop their own reference library and search tools.

Reference Libraries and Matching Software

Reference libraries are the backbone of any spectral matching software. The use of spectral libraries in microplastic research has increased from 33% in 2015 to 62% in 2017.[70] Commercial spectral libraries host one or more types of reference spectra (e.g., National Institute of Standards and Technology, NIST; KnowItAll, Bio-Rad; Polymer Spectral Libraries, Thermo Fischer Scientific, and S.T. Japan). Although useful, these libraries are only available via paid subscription. This is a costly commitment, limiting usage to those with larger research budgets. Commercial libraries also tend to be composed solely of spectra from pure polymer reference materials, whereas particles found in the environment often include additives (e.g., dyes, plasticizers) and have experienced weathering and/or degradation. Because of the high cost and limited scope of commercial libraries, 30 of 53 papers that used spectroscopy in our review created their own collections of reference spectra specifically for microplastics.[9,51,55,59,60] Of these papers, many did not provide clear access to their in-house spectral library within the manuscript or supplementary materials, and therefore the use of these reference spectra is restricted to a singular study or research group. Very few plastic libraries have been shared to date, including one for use with Raman spectroscopy[46] and one for IR.[55] There are no freely shared libraries for microplastic analysis using Py-GC–MS or EDS. Shared spectral libraries are of great benefit to

all within the field because they save new users the time required to collect materials and develop their own libraries. Sharing libraries also allows for more reliable matching, as more reference spectra increases the lines of evidence to make a conclusion with. In contrast, screening for potential plastic contaminants in a given laboratory is one case where it will always be necessary to develop an internal library.[27] Although a complete spectral library of all possible contaminants will not be possible in most cases, the spectral fingerprint of some possible sample contaminants can improve post-processing for contaminant exclusion.

Library Matching Procedures

Although polymer libraries exist for the accurate classification of materials, issues still arise in the matching process. Microplastics come in many different forms, colors, and with a variety of chemical additives.[14] As a result the spectra characteristic of one polymer type can differ widely, depending on the particle's specific chemical mixture. Spectra from different points on the same particle can be extremely different. Plastic identification can also be complicated by polymer composite laminates. Furthermore, polymers from environmental samples are often exposed to conditions such as photo degradation, thermal degradation, and biodegradation,[117] which can cause leaching or denaturing of chemical components, altering the original polymer composition and thus its chemical fingerprint.[51] Signals from dyes, additives, and organic matter can also cause interference and limit successful polymer classification. Strategies for spectral matching are diverse, as are the matching algorithm and threshold used to distinguish between a positive match and a non-detect. We review these factors as they relate to the methods of IR and Raman, Py-GC–MS, and EDS separately since each technique has drastically different terminology and considerations.

In IR and Raman spectroscopy, the key considerations for spectral matching are an equal spectral range, the threshold used for positive classification, and the applied algorithm. The choice of spectral range to match is a serious consideration with some studies matching the whole spectral range (0–4000)[29] and others only a subset such as 800–2200 cm^{-1}.[68] If the peaks of interest are not represented in the range used, a spectrum cannot be matched. Comnea-Stancu et al.[68] found that the reduced range 800–2200 cm^{-1} was able to differentiate plastic from natural material since the highest density of plastic peaks are represented within the range. The percent match is another important factor and can be expressed as a hit quality index or an R^2 value.[70] Most reviewed studies count spectra with a percent match greater than 60 to 90%.[29,75,114] For IR and Raman spectroscopy, library searches are normally based on correlation.[70] However, there is a wide diversity of alternative matching procedures to correlation. For example, Zhao et al.[21] constructed their own matching procedure in R based on Euclidean similarity matrix. Lenz et al.[51] manually plotted their spectra first against reference spectra then used the R^2 value as the matching criterion from a correlation between the sample and a reference library. Ossmann et al.[116] used the least squares algorithm to identify matches. Studies have screened the different available matching procedures and found that all run into issues for classification.[9,60] In Primpke et al.,[9] vector normalization combined with spectral correlation yielded the best results. When automating this process, Primpke et al.[9] used two unique matching procedures to validate every match. Primpke et al.[12] and Renner et al.[60] both developed their own open access code for processing and matching IR or Raman spectra. In general, IR and Raman matching methods have been found to perform well when evaluated against known materials.

Polymer classification in Py-GC–MS can be realized by comparisons of pyrograms with references,[118–121] by searching for characteristic peaks in the pyrograms and identifying them with a database (the NIST library or an in-house library)[65,89,122–127] or by averaging the entire mass spectrum and comparing it with a database (a commercial or in-house library).[23,83,128] Each polymer has a different pyrogram and associated mass spectrum allowing distinctions that can be used together to identify the polymer type. Consequently, manual classification of the polymer fingerprint can be performed by comparing the spectra against a reference library. In addition to the fingerprint, classification of characteristic pyrolysis products using a database, such as NIST, can be used to complete manual classification of the fingerprint. Classification using an average mass spectrum has been performed by a few authors.[23,83] Average mass spectrum classification allows the user to fix a minimum similarity threshold between a reference spectrum and the unknown spectrum. For example, Dehaut et al.[83] fixed a minimum similarity percentage of 80%, as recommended by the pyrolyzer supplier. Unfortunately, the software used to conduct these analyses is expensive. Only a few studies clearly mentioned that they used a combination of at least two of the abovementioned procedures for polymer classification.[23,65,126] Hendrickson et al.[65] and Peters et al.[126] used an advanced three-step matching procedure. These authors first used NIST to identify the plastic polymer, second, they compared the mass spectra or the pyrogram to plastics standards in an in-house library, and finally they compared their results with published data. In addition, Hendrickson et al.[65] and Peters et al.[126] also calculated the S/N using Mass Hunter in order to determine the presence of pyrolytic products in unidentified pyrograms. In another work, Hermabessiere et al.[23] first used a software (F-Search) to compare the average mass spectra to the supplier database and their own in-house library. If the similarity percentage was below 80%, the authors compared pyrolytic products with available literature or characteristic compounds

were identified using NIST and linear retention indices. It is worth mentioning that out of the 15 studies using Py-GC–MS for microplastic analysis, 11 used an in-house library or pyrograms from reference/standard polymers.[23,65,83,118–123,126,128] The multitier confirmation approach using pyrogram and mass spectra provides higher confidence in the data quality. One distinct advantage of Py-GC–MS over the spectroscopic methods is that there is negligible effect of shape, density, surface topology, color, brightness, or opacity on the quality of the chemical match.

Energy dispersive spectroscopy provides information on elemental composition. A carbon-dominant peak is tentatively attributed to plastic, since plastics commonly exhibit this strong band. Some organic substances that also contain carbon (C) can be excluded by optical and SEM microscopies based on their biological structures (e.g., parallel striations on shells).[129] Conversely, Anderson et al.[130] created an in-house library of organic materials that they thought could be misidentified as plastic and found a unique band in the Si region that was present in the organic material but not present in the plastic allowing them to differentiate plastic from the contaminant. Two types of plastic polymers have been classified based on chlorine and fluorine, EDS signals in the absence of contaminants as polyvinyl chloride and polytetrafluoroethylene, also known as Teflon, respectively.[95,129] However, real environmental plastic samples with unknown manufacturing and weathering history show the presence of multiple elements such as Al, titanium, sulfur, barium, zinc, calcium, magnesium, sodium, and Si. These can be attributed to (i) plastic additives and pigments added during manufacturing of plastics and/or adsorbed on their surface from the environment, (ii) plastic and/or nonplastic adhered materials, and (iii) false positives such as diatoms, shell fragments, or coal ash.[61,89,93,95,106,129,130] Unfortunately, information gained from EDS is not considered a reliable chemical validation of plastic for most applications.[131] Given the complexity of environmental samples, EDS has to be used in combination with other techniques such as Py-GC–MS, IR spectroscopy, and Raman spectroscopy for a reliable chemical classification of plastic, as described later (Multiple Methods and Data Fusion Techniques section). Moreover, there is no mention of automated spectral matching strategies for EDS spectra, in contrast to IR, Raman, and Py-GC–MS spectra, for which extended libraries and tools exist.[46,74,104]

Good Practices for Developing In-House Reference Library and Matching Tools

Since most of the reviewed studies developed their own reference libraries and matching tools but did not expand upon the considerations involved, we highlight some of the most important aspects in developing a spectral reference library: (i) Keep thorough metadata records. Spectra can differ between labs dependent on their techniques for obtaining clear spectra, and their individual machinery. It is very important to always keep metadata stored with the spectral library (e.g., instrument, manufacturer, laser, instrument settings, processing, and classification parameters). (ii) Create a diverse library. There is a wide range of plastic types, additives, and weathering degradation products, the more of these that are incorporated into the library (as well as lookalikes), the better the library will classify the sample. The likelihood of reliable spectral matching is highly dependent on the comprehensiveness of the spectral library.[132] (iii) Clean library spectra. Modify reference spectra using all processing steps mentioned in this publication to amplify the precision and robustness of the library, e.g., noisy data will only add error to the matching capabilities of the library. (iv) Validate the library. Libraries are not perfect; they may contain errors and even the most advanced libraries need to be calibrated before use. This can be done via cross-validation, or negative control studies.[66] Cross-validation is the simplest method to use, whereby known samples are run against the spectral library and the percentage of correct matches is reported. Additionally, a second step of cross-validation is to run materials through the library that the library is not familiar with and report the match strength of all the incorrect matches. The upper bound of incorrect matches should be used as the quality threshold for false positives and matches below this should not be accepted.[55] Validation should also be done with commercial libraries to determine the hit quality threshold, which may not be specified by the producer. (v) Automate the matching procedure. Automated matching is a requirement for spectral mapping and automation is the key to high throughput techniques. Automated procedures can be written in common coding software (e.g., R, Python, and Matlab) and many instrument vendors offer commercial library matching programs.[9,47,59,60] (vi) Share reference data and matching tools. There are hundreds of additive materials within plastics that have their own characteristic peaks which can create background and baselines within the spectra and many of these additives are proprietary. The only way microplastic researchers will be successful at identifying the full range of plastics is if they join together and share their reference data and matching tools. At time of publication, please share reference library spectra as supplementary materials.

Quality Assurance, Quality Control

The establishment of a quality program to manage (i.e., QA) and maintain (i.e., QC) the accuracy and precision of research activities is essential to any environmental monitoring program.[133] QA/QC is described thoroughly in Brander et al.[79] within this special issue. Here, we focus on QA/QC of analysis procedures, which has been conducted in some microplastic studies.[9,46,55,60,134–136]

The most commonly used analysis validation techniques for visual and spectral procedures were blank[110,137,138] and positive controls.[139–141] Visual QA/QC techniques are currently more qualitative and manual than the quantitative and automated QA/QC procedures used in spectral matching. Some visual analysis studies discussed validating their visual classification by having multiple observers discuss any uncertain classifications.[87] High-quality photo imagery can help to further QA techniques by facilitating sharing with researchers from other labs for added assurance. Blank controls are used to calculate false positive rate, while positive controls can be used to calculate false negative rate. Positive controls should always be used when developing new methods and are good practice for general calibration of methods throughout studies (see Good Practices for Developing in-house Reference Library and Matching Tools section). Spectral classification techniques are also considered a validation step for visual classification.

Opportunities for QA/QC

Opportunities for QA/QC development include double blind tests, negative controls, reporting, and data sharing. Double blind tests could help to decrease bias in training regimens or while testing analytical procedures that require observer tuning parameters such as baseline correction and smoothing in Raman and IR spectroscopy. Negative controls could consist of adding lookalike materials to plastic to spike recovery experiments. These could help to refine calculation of false positive uncertainty. QA and QC procedures that are used to validate instruments, analytical techniques, and chemometric results should be reported at time of publication and described. Lastly, sharing the source code or spreadsheet for analysis at the time of publication would address many of the reproducibility issues that may result from a lack of analysis validation.

Data Storage and Sharing

In the age of information, data storage and sharing are important. Data storage is the consideration of the security and size of the files that are created (e.g., megabytes, MB; gigabytes, GB). There is a threat of data loss from inadequate data backup procedures, which becomes more complicated and costly as data file sizes increase. Data sharing is the process of freely providing information for the benefit of the scientific community. Data sharing is essential for reproducing, comparing, expanding methods. For example, large amounts of data are necessary to create methods using machine learning that can accurately count, characterize, and classify microplastics. As the analytical capabilities of microplastic researchers expand, new platforms for data need to be implemented to assist in data storage and sharing.

Storage

Few studies report the size of the data set they create but the size of data can be a defining factor of whether or not other labs have the capacity to reproduce the analytical method. For example, Peeken et al.[134] reported that they collected 112.5 million spectra using spectral mapping from an focal plane array (FPA)-based FT-IR system that created 627 GB of information per sample, while Käppler et al.[142] reported that they analyzed the spectra of 27 particles which we could infer to be on the order of MB to a GB of data. These two studies are sharply contrasted by the size of their data sets. To replicate the study of Peeken et al.,[134] one would need to have access to very large storage capacity, memory, and computing power.

Sharing

Of the reviewed studies, none shared data and analysis code completely in line with all FAIR[143] (Findable, Accessible, Interoperable, and Reusable) and Open data principles.[144] However, the authors of one report shared all data necessary to reproduce the analysis of the study (assuming access to proprietary software Primer v.6) including large files for FT-IR imaging data as reference data sets.[55] Four studies shared source code for analysis.[9,55,59,60] Less than 50% of the reviewed papers shared any raw data in a manner that could be used to reanalyze and reproduce at least a portion of their results. Two publications shared in-house spectral libraries.[9,46] The most common type of data that was shared in publications were images of plastics that had been visually identified.[73,145,146]

Opportunities for Data Storage and Sharing

Currently, data sharing is dramatically insufficient. Sharing raw data and supporting analysis code is the gold standard of reporting scientific results and analysis methods. The large size of the spectral databases created during microplastic investigations pose new challenges for data reporting, sharing, and analysis time. Data sets of hundreds of GB to terabyte size cannot be easily shared on publication platforms or hosted on a downloadable server at this time.[134] Costs to the host are too great. However, Käppler et al.[142] could easily share their MB size data set on many online data portals free of charge. However, every attempt should be made to make large data sets available on request. There are very user-friendly and freely available data and code sharing platforms (e.g., Figshare, Google Drive,[12] Open Specy,[71] and those available from the journal publishers). When R, Python, or Excel are used, there should be a trail of analysis stored in the source code or spreadsheet. These codes and spreadsheets should be shared for quick reference and reuse of the analysis tools. This will save other researchers countless hours developing their own.

As a template for data sharing, the A. Gray research group at UC Riverside shared images, spectra, and particle dimensions from some of their data to the Open Science Framework (OSF; Fig. 6).[92]

Future Outlook for High Throughput and Automated Techniques

As microplastic research is rapidly creating new publications, chemical identification is becoming a more common method of choice, and the number of methods used in publications is increasing (Fig. 10). The future of microplastic research will require high throughput automated techniques. These techniques will likely include a chemical identification step and will utilize a diversity of methods as evidenced by the increase in the number of methods employed. While chemical identification has been automated using the FT-IR focal plane array, Raman spectral mapping and particle finding capabilities, these techniques are not high throughput,[9] and require expensive machinery and computational resources including hundreds of GB of information created per sample.[134] The analysis time for these advanced techniques can take multiple days of computing time.[9] The way forward is to develop novel mixed methods which take advantage of high throughput screening options (e.g., fluorescence microscopy, optical microscopy, SEM) as a primary step for identifying which samples to investigate further with chemical classification methods. These methods may be further amplified in their utility and accuracy by employing machine learning.

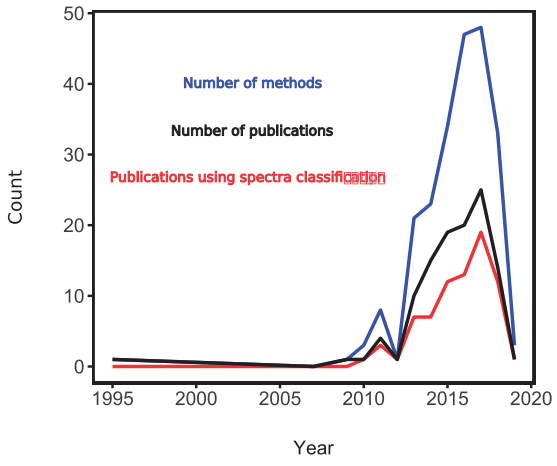


Figure 10. Number of publications in the review (black), number of methods used in the publications (blue), and number of publications that used spectral classification (red). The number of publications increases from 2010 to 2017 and this region of time seems to be the best represented by the data set. Between 2010 and 2017, the average number of methods used per publication increases slightly. Also, the number of papers using spectral classification is tracking the number of publications.

Multiple Methods and Data Fusion Techniques

Multiple methods provide redundancy and accuracy. Most of the reviewed studies employed more than one method (60%). The most common multiple method technique was some form of visual procedure to classify potential plastics combined with some form of spectral procedure to verify the classification and distinguish the polymer type (36%). Interestingly, 15% of the papers employed three or more methods of classification. For example, Wang et al.[129] and Wagner et al.[30] both employed five different classification techniques including optical microscopy, SEM, EDS, IR, and Raman to validate conflicting results regarding plastic and pigment classification. However, there is still uncertainty about how to value conflicting validation results from various procedures. One approach is to discredit the object as a non-plastic if one of the techniques classifies the particle as a non-plastic.[80] This conservative approach could underestimate microplastic risk and abundance when numerous methods are employed and frameworks should be developed for weighing evidence in multiple method approaches.

Future multiple method approaches will hopefully move away from the qualitative nature of current visual classification techniques. Visual or fluorescence-based techniques can serve as a rapid indicator of the potential for microplastic contamination. Once the potential microplastic particles are identified, spectral methods can then be used for further characterization and identification. This implies that visual techniques in the future should be developed to minimize false negatives and spectroscopy techniques should minimize false positives.

Areas that have yet to be explored thoroughly for potential as high throughput screening techniques are SEM or optical microscopy mapping to quantify surface topologies, optical microscopy to quantify colors, darkfield microscopy for identifying transparent microplastics, fluorescence microscopy of undyed samples to look for inherent plastic fluorescence,[11] reflectance microscopy for identifying shiny particles, and the quantification of unique shapes using particle mapping software. These newly quantified variables could be combined as a multispectral image and added to machine learning models to aid in identifying truly unique combinations of features about microplastics.

Machine Learning Applications

To get statistically relevant information on the shape and size of particles in images, segmentation methods are needed that can isolate the particles from the background and from one another. Although automated approaches for detection and counting of objects are well established and widely used in medical image analysis (e.g., to detect cells), these methods are also suitable for the detection of micro- and nanoparticles.[147–151] Traditional segmentation approaches that can be found in literature often use threshold-based methods combined with morphological

operations–which show good segmentation performance for images from different imaging modalities.[147] However, these methods often encounter problems with the classification of irregular object patterns and noise.[152] Additionally, some of these algorithms rely on hand-crafted feature selection or shape assumptions,[150,151] which often limit their use for the detection of well-defined objects with a specific feature within the context of other objects that give rise to contrast in an image.

In addition to traditional threshold-based approaches, machine learning with trainable algorithms can be used to identify particles and interpret images. In recent years, with advancements in machine learning and in particular deep learning,[153] deep convolutional neural networks (CNNs) essentially replace traditional algorithms in various fields, ranging from medical image analysis for different modalities[154–158] to object recognition tasks in general,[159–163] with remarkable improvements in pattern and image recognition accuracy. For example, images of microplastic microspheres and scattered laser light from microspheres have been used in combination with a CNN to find microspheres based on material and shape on a surface.[164] A tool for automated detection of microplastic particles was later developed using these principles.[165] Because of the ability of CNNs to automatically extract task-relevant information from even complex data sets, deep learning networks can achieve performance similar or superior to humans on specialized tasks.[166,167] However, in order to train deep neural networks large-scale data sets with sufficient manually annotated ground truth data is needed, which remains both time-consuming and costly to obtain. To improve this situation, attempts such as "transfer learning" can be applied by using models that are pre-trained on image recognition data sets such as the ImageNet and are subsequently adjusted for the data set specific segmentation task.[168] Another promising approach that requires less data is a method termed "precision learning" introduced by Maier et al.[169] which permits the use of pre-existing knowledge incorporated in neural networks in the form of known operators. This type of approach can reduce the amount of data needed to train a neural network for proper image segmentation, but still requires a certain data set size with manually annotated data.

The machine learning-based detection of micro- and nano-plastic particles in various types of images from optical to electron microscopies is in its infancy. Davaasuren et al.[170] used a machine learning model to interpret synthetic aperture radar image data based on experimentally derived proxies of microplastic hotspots. However, it is unlikely that microplastics will be detectable using satellite-based remote sensing products soon because of low spatial resolution. Lorenzo-Navarro et al.[171] used images of microplastics to train several machine learning algorithms and found the random forest classifier to perform best at discriminating shape factors of microplastics.[172] Bianco et al.[173] used a support vector machine to train a binary microplastics identifier on holographic images.[174] To detect microplastic particles from seawater, Shan et al.[175] introduced a combination of hyperspectral imaging with support vector machines which satisfactorily permits the discrimination of different microplastic types and particles with dimensions between 0.1 mm and 3 mm.

Machine learning was also used to classify microplastics in spectral maps. Hufnagl et al.[176] proposed a random forest approach to detect microplastics in μIR images as shown in Fig. 11. Hufnagl et al.[176] claim that the algorithm

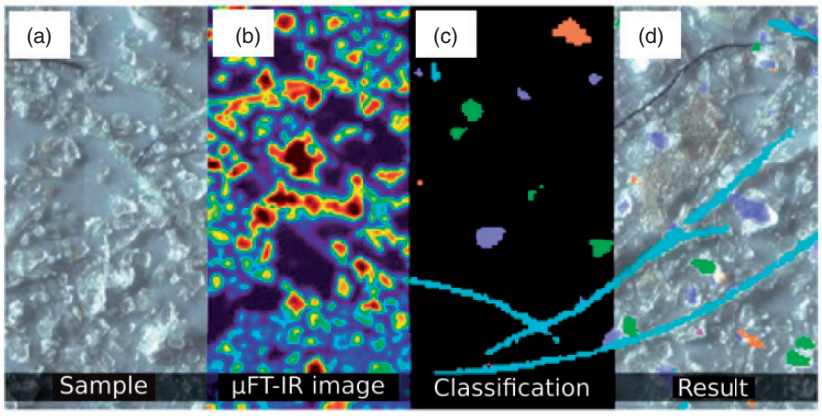


Figure 11. Microplastic classification in μIR images using a machine learning approach introduced by Hufnagl et al.[176] The images from left to the right represent a zoomed-in and panned view of a sample: (a) optical image of the microplastic sample, (b) μIR image, (c) pixelwise classification output of the machine learning model for the different polymer types (including PE, PP, PMMA, PAN, PS, and non-polymer particles), and (d) overlay of the classification result with the optical image of the sample. Reproduced from Hufnagl et al.[176] with permission from The Royal Society of Chemistry.

was able to correctly identify several classes of polymers based on a spectral map with high accuracy. However, the experimental support of this claim is still rather limited (with only six polymer classification classes) even though the approach is promising. Therefore, we conclude that the methods of machine learning are a suitable starting point to substantially advance and automate the discrimination of plastic particles.

The future of machine learning for microplastic classification will likely include multiple methods and data fusion techniques, advanced machine learning approaches, and incorporation of published and shared images. Currently, we could not find any literature that used machine learning to identify plastics using multiple methods (e.g., optical microscopy and IR spectra) but these could be developed and may increase the accuracy of these machine learning models. This review identified 69 publications that share imagery of verified microplastics (Review Metadata, Supplemental Material); we share images from our collection (Images, Supplemental Material),[92] and we wrote a code for downloading images from Elsevier publications using their API (Elsevier API R, Supplemental Material).[91] Here, precision learning and transfer learning approaches have very large potential due to the fact that they exploit and rely on pre-existing data and knowledge and will thus profit from data available today, which continues to grow rapidly.[169]

Conclusion

Research on microplastics is rapidly expanding and is experiencing growing pains that need to be addressed to develop high throughput automated techniques. Our major recommendations for opportunities the field should pursue include: (i) the quantification of particles in images by color, shape, size, and surface topology using image analysis software, (ii) use the quantified values to identify threshold values in images that differentiate plastic particles from other particles, (iii) further develop spectral data processing using robust techniques developed specifically for plastic, (iv) follow best practices for creating and sharing robust spectral libraries, (v) improve QA/QC by conducting double blind and negative controls, (vi) share raw data and analysis code with publications, and (vii) leverage readily available data to develop machine learning models. Microplastic research has come a long way in a short time, but there is still so much work to be done.

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Supplemental material

The supplemental material mentioned in the text, consisting of the review methods, is available in the online version of the journal.

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